RiskDrive: Multimodal-Based Risk Map Prediction for Autonomous Driving

Runhao Li
1006746309
rainli@mail.utoronto.ca
`https://github.com/rainli10/riskmap`

1 Project introduction

Autonomous driving scenes contain many objects, but not all regions are equally important for safe navigation. Nearby pedestrians, vehicles, and cyclists are much more critical than distant background regions, so an autonomous system should understand not only what is in the scene, but also how risky each region is. In this project, I predict a dense 5-class per-pixel risk map from RGB-D input, where higher values indicate more dangerous regions for the ego vehicle. To address this task, I propose a SegFormer-based architecture adapted for RGB-D input, which combines pretrained semantic feature extraction with depth-aware scene understanding for per-pixel prediction. Compared with a simple CNN baseline trained from scratch, the method achieves substantially better results.

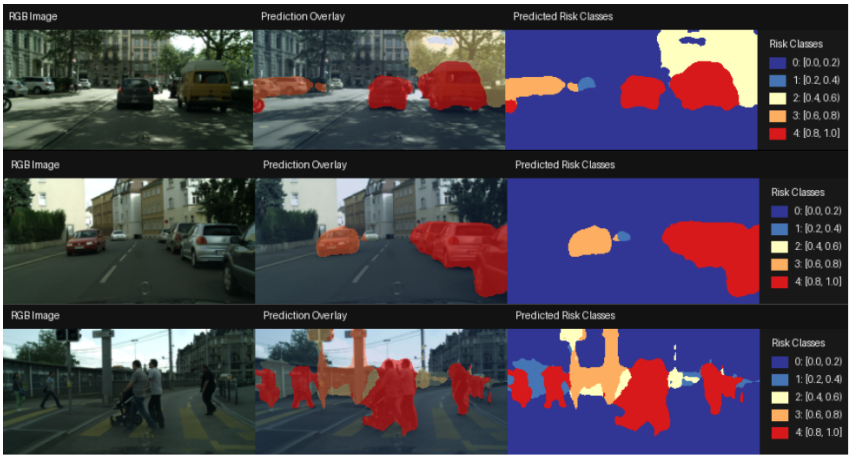

Figure 1: Predicted Risk map for autonomous scene

2 Background & Related Work

The Cityscapes and KITTI datasets provide widely used benchmarks for autonomous driving, offering high-quality semantic annotations and geometric information that support pixel-level information in urban environments [3] [4]. Our task of predicting risk maps is closely related to semantic segmentation by extending segmentation from learning semantics to level of risk with integrating information from depth input. To provide geometric information, we use Depth Anything [8] to generate depth maps from the RGB images, which supply reliable proximity cues that help the model better estimate scene risk, especially for distinguishing nearby dangerous objects from farther low-risk regions. Risk map prediction is also object oriented which share the same pipeline as segmentation models do. Early semantic segmentation for autonomous driving was dominated by encoder–decoder architectures such as U-Net shape. DeepLabv3+ builds on this paradigm by combining hierarchical feature extraction with spatially precise decoding, achieving strong performance on benchmarks like Cityscapes [1]. More recently, transformer-based models have gained prominence: SegFormer replaces convolutional encoders with hierarchical transformers for global context modeling, while

Mask2Former introduces a unified transformer framework for dense segmentation and reaches state-of-the-art performance on Cityscapes [7] [2]. More recently, Vision Transformer (ViT)–based segmentation models have gained attention for their ability to capture long-range global context, which is particularly beneficial for dense prediction tasks in complex urban scenes, as demonstrated by models such as the Segment Anything Model [5].

3 Data Processing

I built the risk-map dataset from the Cityscapes [3] driving dataset by using RGB image together with its semantic segmentation map and a generated depth map from DepthAnything Model [8]. I constructed a new supervision target by combining semantic importance and geometric proximity. Specifically, each semantic class is assigned a fixed risk weight w_c based on its expected danger level in road scenes. We then convert depth into a normalized proximity weight using

$$g(D) = \text{clip}\left(\frac{1/D - 1/D_{\max}}{1/D_{\max} - 1/D_{\min}},\ 0,\ 1\right)$$

so that closer objects receive higher values. The final dense risk map used as supervision is computed per pixel by

$$R(x, y) = w_{c(x,y)} \cdot g(D(x, y))$$

The example of this risk map is showing in Figure 2. This dataset design was one of the main challenges of the project, since risk map prediction is not a standard benchmark task, so we had to define the labeling logic ourselves based on everyday driving intuition, including the semantic weights, inverse-depth weighting, and overall risk construction process.

(a) RGB (b) Semantics (c) Risk Map Discrete (d) Risk Map Continuous

Figure 2: Example training sample used in our dataset construction.

4 Architecture

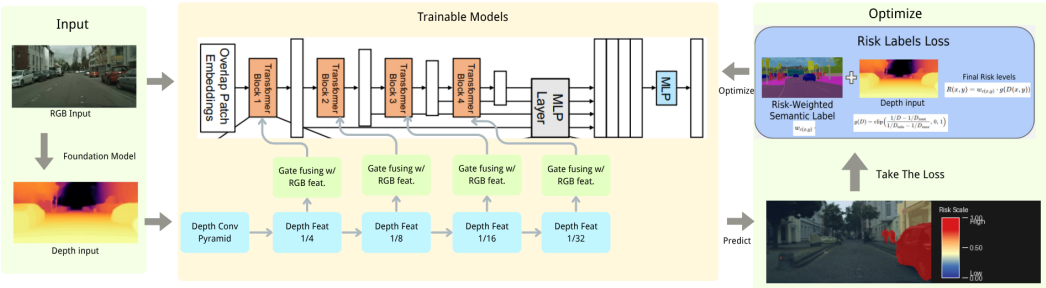

Figure 3: Architecture of RiskDrive

Our final model is a depth-gated SegFormer for dense 5-class risk prediction from RGB-D input. We first generate a depth map from the RGB image using a foundation depth model. The RGB image is processed by the SegFormer backbone, while the depth map is passed through a lightweight convolutional pyramid to produce multi-scale depth features at matching resolutions. At each stage, a learned gating module fuses the depth feature with the corresponding RGB feature before it is passed into the transformer blocks, allowing geometric proximity cues to influence feature extraction throughout the backbone. The final multi-scale features are then aggregated by the standard SegFormer MLP decoder to produce a dense per-pixel risk map. This design allows the model to combine semantic scene understanding from RGB with distance-aware information from depth, which is critical for accurate driving risk estimation.

We train the depth-gated SegFormer in two stages. In Stage 1 (epochs 1–8), the pretrained SegFormer backbone is frozen, and only the newly added depth encoder, gating modules, and decode head are trained using AdamW with cosine annealing. In Stage 2 (epochs 9–30), the backbone is unfrozen and the full model is fine-tuned, using a smaller learning rate for the pretrained backbone and a larger learning rate for the newly added layers. In both stages, gradient clipping and automatic mixed precision on CUDA are used for stable and efficient training.

Baseline Model

As a baseline, we use a simple CNN trained from scratch for dense per-pixel risk classification. The model takes a 4-channel RGB-D input and outputs a 5-class risk map at the same spatial resolution. The architecture is fully convolutional, with six Conv-BatchNorm-ReLU blocks of channel sizes $32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 64 \rightarrow 32$, followed by a final 1×1 convolution layer for per-pixel classification. The baseline is trained with class-weighted cross-entropy loss to address the imbalance between common low risk pixels and much rarer high risk pixels.

5 Results

Table 1: Comparison between the baseline CNN and our model on the test set.

Model	Validation Loss ↓	Pixel Accuracy ↑	MAE in Risk Space ↓
Baseline CNN	0.0933	0.809	0.0640
Our Model	0.0523	0.914	0.0396

5.1 Qualitative Results

Qualitatively, our model produces coherent risk regions with clear separation between important objects and surrounding background. It identifies pedestrians, vehicles, and other critical scene elements reliably, while capturing their surrounding risk context in a consistent way. However, the predicted boundaries are not always fully precise, and some object edges still appear slightly blurry.

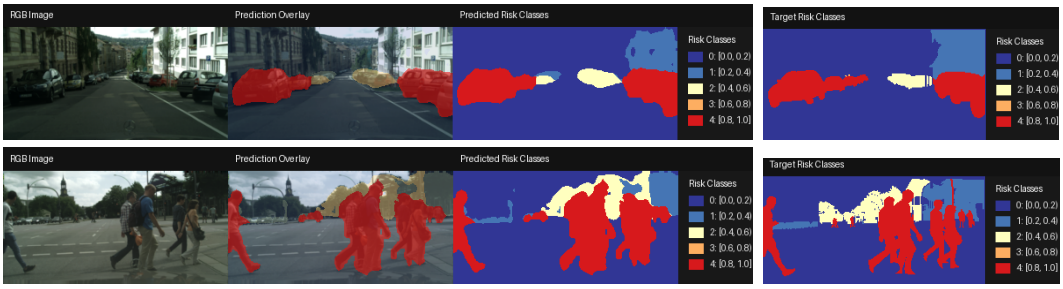

Figure 4: Qualitative comparison of predicted risk maps and ground-truth risk maps.

5.2 Quantitative Results

Our model consistently outperforms the baseline CNN on all three evaluation metrics. It reduces the validation loss from 0.0933 to 0.0523, improves pixel accuracy from 0.809 to 0.914, and lowers the MAE in risk space from 0.0640 to 0.0396. These results indicate that the proposed depth-gated SegFormer produces more accurate per-pixel risk predictions and better captures the semantic and geometric cues required for dense risk estimation.

5.3 Performance on New Data

To evaluate generalization on new data, we tested our model on KITTI-360, an unseen dataset that is different from the Cityscapes data used for training. We selected KITTI-360 because it provides 2D semantic annotations, which are necessary for constructing reliable ground-truth risk maps for metrics evaluation. If we uses inferred semantic labels would introduce additional uncertainty into the evaluation target. On this unseen dataset, the model achieved a loss of **0.0581**, pixel accuracy of **0.923**, and MAE of **0.0419**, which are very close and in the case of pixel accuracy slightly better than the performance on our previous test set. This suggests that the model generalizes well across datasets and does not overfit heavily to the visual style of Cityscapes. A likely reason is that the model learns scene-level semantic and geometric cues that transfer across urban driving environments.

Figure 5: Qualitative analysis on KITTI-360 and DVP highway in Toronto.

Qualitatively, the KITTI-360 [6] results are also strong: in our examples, small pedestrians and vehicles are correctly identified and assigned appropriate risk regions. We additionally tested the model on self-collected driving data from Don Valley Parkway in Toronto in Figure 5. Since this data does not have ground-truth semantic labels, we could not evaluate it quantitatively, but the qualitative results are encouraging. In particular, both nearby and distant vehicles were detected with reasonable risk estimates. However, the same limitation observed on Cityscapes and KITTI-360 remains: object boundaries are not always sharply segmented, and some risk regions still appear boxy or slightly blurry. Overall, these experiments suggest that the model shows stronger than expected generalization to unseen real world driving scenes and satisfy real2sim transferring.

6 Discussion

Our results show that the proposed model performs well and generalizes better than expected. On the original dataset, it achieves strong quantitative performance, and this performance remains consistent on the unseen KITTI-360 dataset where the metrics stay very close to the original test results. This suggests that the model is learning transferable semantic and geometric cues rather than overfitting to the visual appearance of Cityscapes. The qualitative results on real-world driving data from Toronto further support this conclusion: nearby and distant vehicles are assigned reasonable risk levels even without dataset-specific tuning. The most interesting finding is that the model shows strong cross-dataset and sim-to-real robustness, which indicates that combining transformer-based scene understanding with depth-guided fusion is effective for dense risk estimation. At the same time, the remaining boundary blur suggests that the main limitation is no longer recognizing risky objects, but producing sharper and more precise object-level segmentation.

Ethical Considerations This project uses public autonomous driving datasets that already apply anonymization procedures such as blurring faces and license plates, but privacy risks may still remain if this process fails in rare cases. In addition, our risk maps are generated from heuristic assumptions about semantic danger and object proximity, so errors or overconfident predictions could misrepresent real driving risk and lead to unsafe downstream decisions.

Project Difficulty This project is particularly challenging because the task itself is highly novel. There are no standard benchmarks, no well-established baselines, and no guidance on how to approach dense risk map prediction. As a result, one of the hardest parts at the beginning was simply deciding where to start. Before obtaining a working system, we explored multiple architectures to determine a model design that could produce meaningful dense risk predictions. We also had to build a new dataset pipeline and design a novel method for constructing risk-level maps, since no existing dataset directly provides this form of supervision. In addition, even the problem setup was nontrivial, including deciding how risk should be represented and which evaluation metrics would be appropriate for such a new task. Because this problem has not, to our knowledge, been previously studied in this exact form, much of the work required independent problem formulation rather than straightforward implementation. Developing the final model also required extensive paper reading and careful reasoning, especially in dense segmentation and gated depth-fusion methods, since there was no architecture specifically designed for this exact task to follow directly.

References

[1] Liang-Chieh Chen et al. *Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation*. 2018. arXiv: 1802.02611 [cs.CV]. URL: https://arxiv.org/abs/1802.02611.

[2] Bowen Cheng et al. *Masked-attention Mask Transformer for Universal Image Segmentation*. 2022. arXiv: 2112.01527 [cs.CV]. URL: https://arxiv.org/abs/2112.01527.

[3] Marius Cordts et al. *The Cityscapes Dataset for Semantic Urban Scene Understanding*. 2016. arXiv: 1604.01685 [cs.CV]. URL: https://arxiv.org/abs/1604.01685.

[4] Andreas Geiger, Philip Lenz, and Raquel Urtasun. "Are we ready for Autonomous Driving? The KITTI Vision Benchmark Suite". In: *Conference on Computer Vision and Pattern Recognition (CVPR)*. 2012.

[5] Alexander Kirillov et al. *Segment Anything*. 2023. arXiv: 2304.02643 [cs.CV]. URL: https://arxiv.org/abs/2304.02643.

[6] Yiyi Liao, Jun Xie, and Andreas Geiger. *KITTI-360: A Novel Dataset and Benchmarks for Urban Scene Understanding in 2D and 3D*. 2022. arXiv: 2109.13410 [cs.CV]. URL: https://arxiv.org/abs/2109.13410.

[7] Enze Xie et al. *SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers*. 2021. arXiv: 2105.15203 [cs.CV]. URL: https://arxiv.org/abs/2105.15203.

[8] Lihe Yang et al. *Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data*. 2024. arXiv: 2401.10891 [cs.CV]. URL: https://arxiv.org/abs/2401.10891.